

# “The Problem of Time”

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Translated from the Danish

In his article “Relativitetsprincippet”, pp. 1–30 in the previous volume of “Fysisk Tidsskrift”, Professor Kroman passes an unconditional sentence of condemnation upon Einstein’s so-called theory of relativity, which he characterises as “a decidedly self-contradictory and therefore impossible view”. Presumably many physicists who in and for themselves concur in this judgement will nevertheless be inclined to use the Einsteinian system of formulae, on the consideration that it may be an approximate expression for something correct and possibly lead to new discoveries, even if it is derived from an incorrect conception. Whether the danger there is in working with formulae whose real meaning is not understood can be outweighed by the advantages the formulae offer can presumably be known only by the fruits; but the danger is undoubtedly diminished by the fact that one can take refuge in the Lorentzian interpretation of the phenomena. Little satisfactory as this may seem, it may nevertheless be a good makeshift so long as the contradictions which have occasioned the appearance of Einstein’s theory have not been got out of the world in some other way.

However the journal’s readers may now view these things, I believe that the attack which has been made from the Einsteinian side upon our concepts of time may make it desirable that the problem of time should be taken up for closer consideration with a view to Einstein’s theory. In what follows an attempt in this direction shall be made. It cannot be avoided that one thing and another which has already been said in Prof. Kroman’s article is thereby repeated; but that article nevertheless hardly makes superfluous an exposition which aims more directly at throwing light upon our concepts of time.

Under the domain of the problem of time there belong two essentially different kinds of task, namely 1) *the ordering of points of time in temporal sequence* and 2) *the measurement of lengths of time in units of time*; they will here in the main be treated each by itself.

As regards the ordering of points of time, the matter is to determine a point of time under consideration as past, simultaneous or subsequent in relation to one or more others. One has in this connection generally proceeded on the assumption that before, now and after were “qualities in our consciousness”, fundamental concepts in our view of the world, fundamental forms of apprehension, or whatever one may care to call it, and that it was neither required nor possible to explain their meaning or to define them by something still more fundamental.

After the appearance of the theory of relativity it is, however, emphasised as Einstein’s merit that he has perceived the necessity of giving a definite definition of simultaneity between events at different places in space, and has moreover made the remarkable discovery that it was impossible to make such a definition unambiguous. The truth would rather seem to be that the concept of simultaneity which follows from the nature of our consciousness in reality entered earlier, without definition, in the right way into the physicists’ trains of thought, while Einstein’s definition has created confusion, because it has brought it about that limitations in our physical means of determining simultaneity have been exalted into something fundamentally essential for our knowledge. But since doubt has been aroused in this matter, it may be useful to make quite clear to oneself what is the real content of the concept of simultaneity (for events at different places) hitherto used without closer explanation, and this is at bottom not difficult.

Let us think of two places  $A$  and  $B$  — which may be in motion or at rest in relation to each other. From  $A$  there is sent out, at a point of time to which we may give the designation  $t_1$ , a messenger to  $B$ , where it arrives at a for the present unknown point of time with the designation  $t$  (in  $A$ -clock-time). The messenger is sent from  $B$  straight back to  $A$  and arrives there at the point of time  $t_2$ . We shall then for the present establish that  $t$  lies between  $t_1$  and  $t_2$ . The shorter this interval of time is, the greater is the average velocity with which the messenger has traversed the outward and the return path, provided these are straight lines. Properly, certain reservations should here be made (for an oscillating motion of  $B$ ); but they lose their significance when the interval of time  $t_1 t_2$  becomes sufficiently small. If we now think of the messenger’s velocity as increasing indefinitely, while the interval  $t_1 t_2$  decreases towards zero, or in other words that the two events, the messenger’s dispatch and its return, approach indefinitely to simultaneity, the point of time  $t$  for the messenger’s arrival at  $B$  will also approach indefinitely to simultaneity with  $t_1$  as well as with  $t_2$ .

One may give what has here been developed the following more definitional expression: two events at the places  $A$  and  $B$  are simultaneous when a dispatch of a messenger,

*however fast the messengers used may be*, can be arranged in such a way that a messenger from *A* arrives at *B* and departs again from there simultaneously with the “*B*-event”, while the point of time for the “*A*-event” lies between the points of time for the messenger’s departure from *A* and its return.

One might ask whether by this definition of simultaneity at two places full reciprocity is attained, that is to say, whether an event at *B* which has been found to be simultaneous with an event at *A* by a dispatch from the last-named place would also be found simultaneous with it if the dispatch went out from *B*; since one of the messenger’s journeys did after all set out from *B* at the very moment the *B*-event took place, this cannot be doubtful. — We can also connect events at three or more places in space in time by letting the messenger take the way from *A* back to *A* round by them all.

One might object that we are not able to send any messenger with infinitely great velocity, or even with a velocity so great that the interval of time  $t_1 t_2$  always becomes of vanishing magnitude in relation to our usual measures of time: our fastest messenger, the light ray, takes years to traverse the way between the earth and the nearest fixed star. This objection, however, does not touch the core of the matter at all. What is in question here is not at all how exactly we can carry out a determination of simultaneity with the means at our disposal, but rather what lies in the concept of simultaneity. With regard to this there has not previously been any disagreement between physicists and other people. When a man, in order to determine the point of time of an event, looks at a clock and establishes that it happened at such and such an hour by the clock’s indication, that is to say, that it occurred simultaneously with the hand’s passage over a definite mark on the dial, he thereby proceeds, consciously or unconsciously, on the assumption that light rays, nerve currents and the like have been messengers so fast, the ways they had to traverse so short, and therefore the whole time for the dispatch so slight, that the deviation from the ideal simultaneity determined by infinitely fast messengers is without significance for his purpose; (Einstein too must presumably, when he speaks without preceding definition of simultaneity between positions of clock hands at *A* and events in “*A*’s immediate surroundings”, proceed on the same presupposition). And when the physicist compared points of time under conditions where the time for the dispatch could not be regarded as vanishing, he took this time into the reckoning as well and as exactly as was possible and necessary for the purpose, and thereby proceeded on a concept of simultaneity agreeing with the definition on p. 3.

The definition of simultaneity between points of time at different places without relative motion which Einstein set up in 1905 did not in and for itself stand in conflict of

principle with the aforementioned definition either, since the latter would give the same result  $t = \frac{1}{2}(t_1 + t_2)$ , provided the presupposition made by Einstein, that light took an equally long time on the outward and the return path, was correct. But since his dogma of the constancy of the velocity of light led to results which stood in conflict with the ordinary concept of simultaneity, this had, like other refractory concepts, to give way to the dogma, and then there was proclaimed — by Minkowski and others — “a complete revolution of our ideas of space and time” as an epoch-making result of exact physical measurement and exact mathematical thinking.

It was immediately evident that a velocity greater than that of light would be fateful for the whole structure of thought, especially for its paradoxes of simultaneity. It was doubtless felt that the circumstance that the velocity of light is the greatest velocity hitherto known was far too weak a defence for a theory which went straight at our fundamental concepts. One believed, however, that one could go a step further and declare the velocity of light to be the greatest physically possible velocity; in support of this it is adduced that a moving body’s energy becomes infinite at a velocity like that of light; but even if this could be regarded as proved, it is not thereby excluded that a velocity of propagation (gravitation’s?) might exceed that of light. Real significance for our conception of time this question has not, however; for even if the velocity of light occupies a special position in the physical respect, it does not occupy a special position for thought. One might say that we know very well a far greater velocity, namely that with which thought is able to fly to the outermost limits of the Milky Way in a vanishing moment. Since one or another might have misgivings about thus placing side by side a mental and a physical velocity, it will perhaps nevertheless be better to keep to the mode of expression that we can think of velocities a thousand, a hundred thousand and any number whatever of times as great as that of light, and that we are entitled to consider Einstein’s theory, like everything else in the world, in the light of such a thought-experiment.

This test in thought is, however, not the only one that can be applied. We can also take our starting point in the determination of simultaneity by clocks carried along, which plays so great a part here on earth. In principle it is quite independent of the velocity of light, and it has as its only presupposition that every change in the course of some process or other, for example a change in a clock’s rate, must have its definite and sufficient physical cause. , If two clocks are absolutely alike in build, and are at a definite place set alike to begin with, they will, subjected to the same physical conditions, constantly keep together. This agreement will not cease because the one clock is moved

to another place, in so far as it is not thereby exposed to other physical influences (that is to say, influences from the material world about it) than those it was subjected to at the place of departure. If we now let one of the two clocks brought into agreement at *A* go along with the messenger to *B* under the method of dispatch spoken of earlier, the messenger would — on the presupposition named — so to speak transport the place *A*'s time-indications with it and make it possible at *B* to set clocks accordingly; as is seen by going to the limit (the infinitely great velocity), one would thereby give simultaneous points of time (by the earlier definition) the same designation at *A* and *B*; there is, in other words, no discrepancy between the two methods of determining simultaneity.

In radium we have perhaps an ideal carried clock, since its degradation seems independent of all external influences; if this really is the case, two pieces of radium, of which the one remained at the place *A* while the other went from *A* on a journey out into the world, should constantly keep together in degradation, so that when at a certain point of time after the separation  $1/1000$  of the one's atoms had exploded, there would simultaneously have exploded  $1/1000$  of the other's. However this may stand, we can at all events think of clocks which were completely independent of, protected against, or compensated for all influence of changes in temperature, pressure, gravitation, magnetic and electric fields or other physical conditions at the point of space where the clock happened at the moment to be. Besides these influences, dependent on the states in the points passed, there might be conceived a physical influence due to the very motion of the clock through a medium, the ether. Such an "ether wind" of unknown magnitude and direction would introduce a hypothetical element into our determinations of simultaneity; but even if we could not get beyond the uncertainty thereby called forth, this would not compel us to alter our ordinary conception of time and space. But if one, like Einstein, does not assume any such ether wind or a corresponding physical influence, an alteration of the clock's rate merely by the motion would, as Prof. Kroman emphasises, be "a causeless phenomenon" — a physical effect without a physical cause. It seems to me that this is quite decisive for the question whether the real alterations in clocks' synchronism (or twins' equality of age!) which are assumed by Einstein's adherents to result from the one clock's (the one person's) journey in space are acceptable from a physical standpoint. Just as little as the clocks' rate can the measuring rods' length be altered by the motion; but when one, by carrying a measuring rod that has been compared in France with the right standard metre "on board" other moved systems *B*, *C* . . . , makes it possible in these to produce right metres, Einstein's ingenious method for comparing measuring rods (the setting of marks during the passage at supposedly simultaneous

points of time) becomes superfluous. When a clock is transferred as well, the dispute about the points of time also falls away, since indeed the transferred clock's indications, as well as the transferred measuring rod's, must be decisive, and the discrepancies with them show quite plainly that there is something wrong in the measurements which give divergent results, or rather in the Einsteinian presuppositions on which they rest.

Neither the transfer of, nor the Einsteinian experiments with, clocks and measuring rods can be carried out in practice under conditions where the results can reach such exactness that they are of significance for the matter: but the placing side by side of the thought-experiments with the infinitely fast dispatch and the transfer of clocks on the one hand, and the Einsteinian transactions on the other, may contribute to throwing light on the new theory's relation to the natural conception of time. That relation seems to me to signify not a *revolution* of this conception, but a *destruction* of it. If we shake it, then we shake the very foundation upon which we are to build up physics' picture of the world; we are carried out into a condition where a mathematical system may indeed be created, but where the natural control fails, so that absurdities get free play provided only they fit into the system.

We apprehend existence in the form of space and time. As regards time this means that we seek to arrange all events in the whole world in a continuous series of groups whose events are mutually simultaneous, but past or subsequent in relation to other groups. Physics has, however — for the material phenomena — not merely this purely historical task; it is also to find laws which are general, that is to say, independent of the point of time<sup>\*)</sup> and of the place (as such), and by which one may derive the events in any group from those in any group past in relation to it.

There had now, however (through Michelson's experiment), been demonstrated an apparently inexplicable distortion in the picture of the world which the physicists had shaped, in that the earth might seem to be standing still with respect to the ether, thus either to stand still itself or to drag the whole world-ether along with it in all its motions. In order to get over this difficulty Einstein had recourse to the expedient of introducing a multiplicity of world-pictures (as many as there are in the whole world systems of particles with different translational motions), each burdened with its individual distortion. Since this distortion was of quite the same kind in the different world-pictures, which were all declared to be equally good, one got *uniformity* or symmetry as a surrogate for the abandoned *unity*; and even if the individual world-pictures did not admit of being reconciled with one another, it was nevertheless possible to convert

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<sup>\*)</sup>"The difference between one event and another ought not to depend upon a difference in time alone", says Maxwell in "Matter and Motion".

the one into the other by simple formulae; thus once more a surrogate: *convertibility* for *agreement*. Considering that Einstein has thus abandoned the monism which has hitherto always been aimed at in the conception of the material world, and has gone over to a peculiar kind of “polyism”, it sounds almost like a bitter irony when *Emil Cohn* in his little work “Physikalisches über Raum und Zeit” asserts that by Einstein’s theory greater unity has come into our picture of the world (“... unser Weltbild wird einheitlicher”). Philosophically regarded, his doctrine is to be rejected, because it destroys our conception of the world and puts many mutually irreconcilable – but by simple formulae connected – truths in the place of one. Physically regarded it becomes to be rejected as well, because where one seeks physical causes it offers mathematical transformations instead. We may say that the abandonment of the historical demand that all events be arranged in a universally valid temporal sequence entails the abandonment of the physical demand that causal connection be established within the material world.

For the illustration of this side of the matter there shall here be considered the case, brought forward by Prof. Kroman as well, concerning the velocity in flowing water.

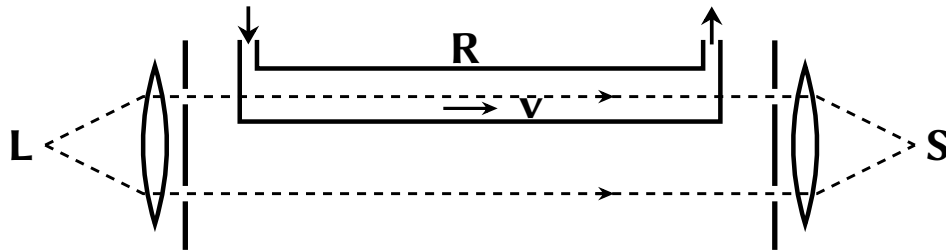


Fig. 1.

In Fig. 1 Fizeau’s experiment is indicated in simplified form. From the light source *L* a bundle of light rays is sent through the air, another through the water-filled tube *R* to *S*, where interference fringes are formed, and these are displaced when the water is made to flow through the tube with the velocity *v* in the direction of the arrow. As is well known one finds a displacement corresponding to this, that the light’s velocity in the stretch of tube is altered by the flow from  $\frac{c}{n}$  to  $h = \frac{c}{n} + \frac{n^2 - 1}{n^2} v$  in relation to the tube wall, and thus to  $h' = \frac{c}{n} - \frac{v}{n^2}$  in relation to the mass of water. There then arises the question why the light gets precisely this velocity, instead of, for example, preserving an unaltered velocity either in relation to the water or in relation to the tube wall.

The physicist means by this question that he wants a closer account of the matter. He wants clarity about wherein the physical processes which take place in the tube when the water flows differ from those which take place when it is at rest. He will next seek either to bring a previously known and more general rule to bear on the case in hand, or

to set up a new plausible theory which can be applied both to this case and under other conditions. But already to penetrate into the nature of the processes, or merely to have one's eye opened to certain possibilities in this respect, is a real advance on the physicist's road of work. One might, for example, make the quite crude assumption that the light passes alternately through the atoms of the substance (or condensed ether-spheres about them) with a certain velocity  $c_1$ , and through the free space between them with the velocity  $c$ . In the flow the atoms are carried along with the velocity  $v$ , and it might then be conjectured that the light preserved its velocity in the empty spaces as well as inside the atoms (in relation to these); at a certain ratio between the "atomic diameters" and the "free paths" one would get the result the experiment gives. But it might also be thought that the velocity of propagation itself inside the atoms was altered to  $c'_1$ , because the light met the atoms with another (relative) velocity. Or one might assume that the light can slip through the interspaces of the atoms, but that its velocity at rest is reduced, by the action of the atoms' "force fields", to  $\frac{1}{n}$  of the velocity in free space; the atoms' motion (in relation to the ether?) would then alter this influence, whereby the light can get another velocity in relation to the tube wall as well as in relation to the water. Or it might be that the velocity with which the front of a wave advanced was not altered by the water's motion, but that the electrons in the mass of water acted upon the phase of the light motion in such a way that "the phase's velocity", which is of course determinative for the interference, was altered;  $h$  and  $h'$  would then denote the phase's velocity. (Which of these or of other possible assumptions is the right one plays no part in this connection.)

The Einsteinian treatment of the question is, however, of quite another kind. By the application of his transformation formulae one finds that a being which goes along in the water's translation should, by any measurements it might make, come to the result that the light motion inside the water has, in relation to this, not the velocity  $\frac{c}{n} - \frac{v}{n^2}$  but rather the velocity  $\frac{c}{n}$ , quite as we find it when the water stands still for us. Even apart from the fact that this is incomprehensible, it cannot be regarded as an explanation. What *I* need is to get coherence into *my* picture of the world; if I cannot get it, it is but a poor consolation that another calls right what I call wrong, especially when I know at the same time that he finds wrong what I find right, so that I should gain nothing by going over to his view of things.

One might say to this that even if Einstein's theory gives no proper explanation of the matter, one could nevertheless in the case considered, by going the opposite way — that is to say, by starting from the simple assumption that for an observer carried along with it the velocity of light in the water is always  $\frac{c}{n}$ , and using Einstein's transformation



formulae — calculate a correct expression for the drag-coefficient which presents itself for observers not carried along with it. This is doubtless true enough; but it is not the question of the use which Einstein's system of formulae may possibly be of that is to be discussed here. There might however, particularly on the occasion of Prof. Kroman's remark on p. 24, be reason for this special case to emphasise that the drag-coefficient  $\frac{n^2 - 1}{n^2}$  can also be derived without the help of the Einsteinian formulae, for example by setting up the condition that light which is led forward through a stretch of standing water and back through a stretch of air, and light which traverses the same way in the opposite direction, shall get no demonstrable interference-producing path difference from the motion of the experimental apparatus through the ether (with the earth); since now Einstein's formulae are constructed precisely so that the "ether wind's" influence shall by them disappear from the calculations as it does from the experiments, it was presumably really given in advance that in the case considered they would give the same result as one can get by directly writing down the conditions for the ether wind's having no influence.

As we now pass over to the other part of the domain of the problem of time, namely the measurement of lengths of time, we ought to notice that nowhere in what has been said above about the determination of simultaneity has any use been made of the concept of uniform motion. Of the clocks (p. 5) it was merely presupposed that they, when they were quite alike and were exposed to quite the same physical influences, would constantly keep together; whether they go uniformly is quite another matter. This notwithstanding, one can by means of the clocks compare lengths of time at different *places*, in that two lengths of time which lie between two sets of two-by-two simultaneous points of time for events at two places are set as equally long. But we cannot, in the nature of the case, get any sure criterion that two lengths of time which elapse at different *times* are equally long. One can only compare such lengths of time by "simultaneity-conferring" them with a time-measurer which goes uniformly; but that this is the case must again be established by comparison with another time-measurer, and so on.

One might get beyond this difficulty by keeping to a time-measurer of which one must, on the basis of one's whole knowledge of the physical world, assume that its rate does not alter in the course of time. The rotating earth is not such a time-measurer, since we can see causes for its period of revolution decreasing. Rather would one presumably acknowledge a measurement of time based on the light motion, taking as an unalterable measure of time the time light takes to traverse a definite length — for example a length equal to the diameter of an electron (disregarding the assumption of

the latter's alterability, or thinking of the closer circumstances as settled). Apparently one might escape mixing lengths into the matter at all, by for example using radium as time-measurer and setting the time in which  $\frac{1}{1000}$  of a piece of radium's atoms exploded as the measure of time; but since the atomic explosions are only clock-strokes for a process running its course in space and time in the interior of the substance, one does not thereby avoid having to do with lengths. The questions of a constant measure of time, a constant measure of length and a constant velocity stand in close mutual connection.

Even if there can be found in nature lengths, intervals of time and velocities which cannot be assumed to be altered in the course of time by any of the physical processes known to us, we cannot nevertheless exclude the possibility that they are altered by some universal cause unknown to us, and in such a way that we cannot check it at all. (The Lorentzian contractions from an ether wind whose velocity and direction are unknown to us afford precisely an example of uncheckable alterations.) Let us assume that the whole world, or at all events that part of it which we know, were subjected to a continued process of shrinkage whereby all lengths were diminished in the same ratio, so that, for example, both the electrons' dimensions and the distances between them had after a certain number of millions of years gone down to  $\frac{1}{10}$ ; then it would in the first place be impossible to establish this by measurements of length with measuring rods which were themselves diminished in the same ratio. But even if all velocities had remained unaltered, so that all processes now (on account of the shorter distances) ran their course 10 times as fast, the shrinkage could not be established by measurements of time either, since the measure of time also (for example the time light takes to traverse an electron diameter) had become 10 times as small, wherefore one would find the duration of all processes expressed by the same numbers as before. Since all processes in the brain and the nerves ran their course with tenfold velocity, one would presumably have to assume that the same would hold of our whole course of ideas, and that our time-sense, that is to say our immediate apprehension of lengths of time, would undergo such a change that the "shrinkage" would not make itself noticed in our consciousness at all. But even if we think of our time-sense as remaining unaffected by the changes, so that we noticed that all processes went on with a far greater speed than before, we could nevertheless not decide whether this was due to the world's having shrunk, or to the world-machinery's having really run wild — or finally merely to a mental change in our conception of time. Just as an increase of all velocities would come to the same thing as a shrinkage of all lengths, a gradual slackening of the world-machinery would act quite like an expansion of all lengths, and none of these four phenomena would be

demonstrable by physical measurements. By a simultaneous shrinkage of all lengths and diminution of all velocities in the same ratio, the duration of all processes would remain unaltered, so that even with an intact time-sense we should not notice any change.

There is evidently in all this an abundance of “relativity”, and the considerations adduced may perhaps even make it doubtful whether any meaning can be attached to concepts such as the absolute magnitude of a length, of an interval of time and of a velocity. They seem afflicted with an indeterminacy similar to that of the concept of absolute rest in an empty and infinite space. With the concept of simultaneity the case is quite another. In order to attach a definite meaning to two events at different places in space being simultaneous, we do not need to ascribe an absolute magnitude to certain lengths, intervals of time or velocities; it is sufficient that we can think of alterations of them (pp. 2–3). One might object that we can indeed think of a velocity 1000, 10 000 or more times as great as, for example, that of light, but that our thought can nevertheless never reach an infinite velocity, and therefore not “absolute” simultaneity either; although this is in a way right enough, it is nevertheless a sophism, not much better than the famous one about Achilles and the tortoise. It will also be seen that the concepts “absolute simultaneity” and “absolute magnitude”, as they are used here, have nothing analogous about them at all; for in the first expression “absolute” means “completely exact”, in the second “without relation to anything else of the same kind”.

The relativity which is inseparable from our concepts of time does not, like the Einsteinian, come into conflict with our hitherto received conception of time. It does not lead us into a mystical world where space has 4 dimensions. It does not hinder us from remaining in the good old three-dimensional world-space, where 3 and only 3 mutually perpendicular planes can be laid through a point, where simultaneous events are simultaneous, where every physical change has its physical cause, and where two railway trains which run each to its own side with a velocity of 100 km an hour move apart from each other with exactly the double velocity, whether we are passengers on the one train or the other or stand outside and look on.